MODEL QUESTION PAPER

REVISION-2010 Reg. No…………………

SUB CODE:3057 Signature……………….

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/ TECHNOLOGY-NOVEMBER,2011

ELECTRICAL TECHNOLOGY

(Common to AE, BM, EA, EL, EC, EP, MD, TC)

Time – 3 hrs Max Marks-100

Part A

I. Answer the following questions in *one* or *two* sentences

1. What is inductive reactance?
2. What is meant by iron loss in a transformer?
3. State the function of commutator in d c motor.
4. List two applications of stepper motor.
5. What is a Megger? (5x2 =10 marks)

Part- B

(Max Marks – 30)

II Answer any five questions.

1. Define the following (a)active power. (b) reactive power (c) power factor.
2. State and explain Thevenin's theorem.
3. What is meant by armature reaction? What are its effects?
4. Derive torque equation of D C Motor.
5. Explain the principle and operation of synchronous motor.
6. Explain the construction and working of universal motor.
7. Draw and explain DOL starter. (5x6 = 30 marks)

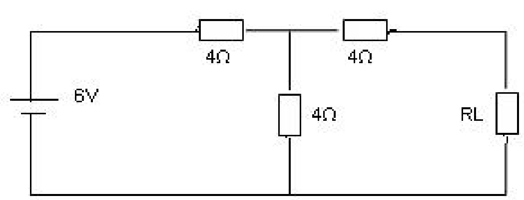
Part-C

(Max Marks- 60)

Answer any four full questions. Select one question from each unit. (15 marks each)

III (a). State and explain Kirchoff's laws. 8

(b) .Find the value of R_L to extract maximum power from the network shown below.



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OR

IV (a) State and explain Super position theorem. 7

(b) Explain the procedure for conducting OC and SC test of transformer. 8

V (a) Explain the working of self excited shunt generator. 7

(b) Draw the diagram of 3 Point starter showing the various parts and state their functions 8

OR

VI (a) What is meant by back emf of d c motor? Explain how the armature current changes when the mechanical load on it changes. 8

(b) Explain the load characteristic of d c series motor. 7

VII (a) With the aid of necessary diagram explain the constructional details of salient pole alternator . 8

(b) Explain working of D C Servo motor. 7

OR

VIII (a) Explain the principle of operation of a 3 phase alternator. 8

(b) Explain the principle of working of reluctance motor. 7

IX (a) Explain the working of a capacitor start induction run single phase induction motor. 8

(b) Explain the procedure of measuring the insulation resistance between. the conductors of an electrical installation. 7

OR

X (a) Why starter is necessary for starting of large induction motor . Explain the Principle of star delta starting of three phase induction motor. 8

(b) With the help of neat diagram explain the construction and working of Megger. 7